Credit Risk Analysis

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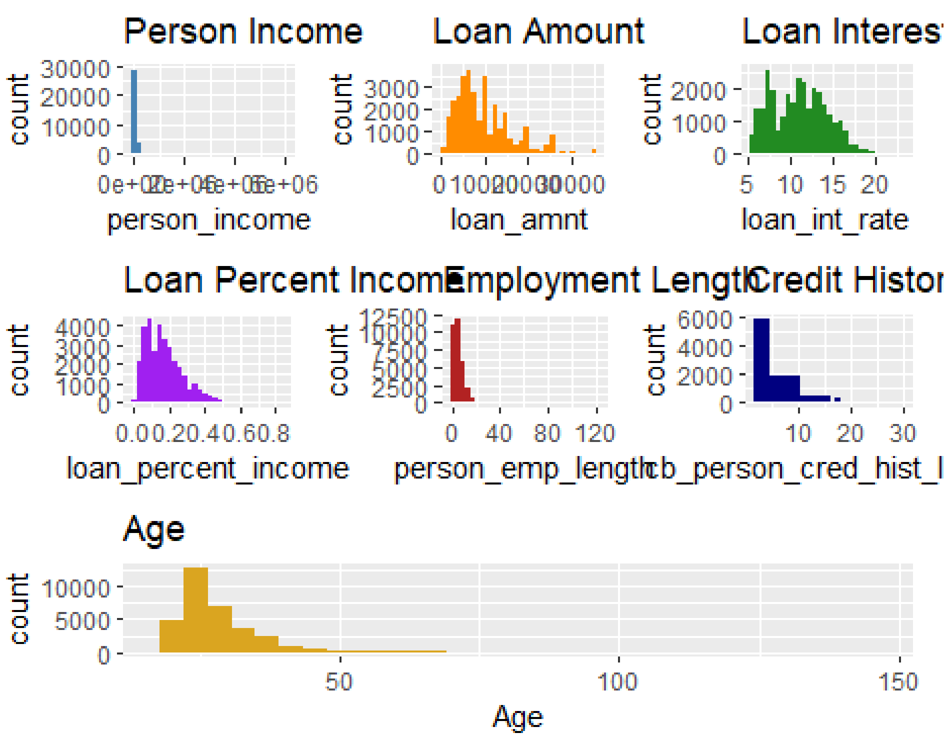
```r
library(readr)
library(ggplot2)
library(patchwork)
library(corrplot)

## corrplot 0.95 loaded

library(fastDummies)
library(xgboost)
library(caret)

## Loading required package: lattice

library(pROC)

## Type 'citation("pROC")' for a citation.

##
## Attaching package: 'pROC'

## The following objects are masked from 'package:stats':
##
##     cov, smooth, var

df <- read_csv("credit_risk_dataset.csv")

## Rows: 32581 Columns: 12
## ── Column specification
────────────────────────────────────────────────────────
## Delimiter: ","
## chr (4): person_home_ownership, loan_intent, loan_grade,
cb_person_default_o...
## dbl (8): person_age, person_income, person_emp_length, loan_amnt,
loan_int_r...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.

dim <- dim(df)
str <- str(df)

## spc_tbl_ [32,581 × 12] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ person_age                : num [1:32581] 22 21 25 23 24 21 26 24 24 21
...
##  $ person_income             : num [1:32581] 59000 9600 9600 65500 54400
```

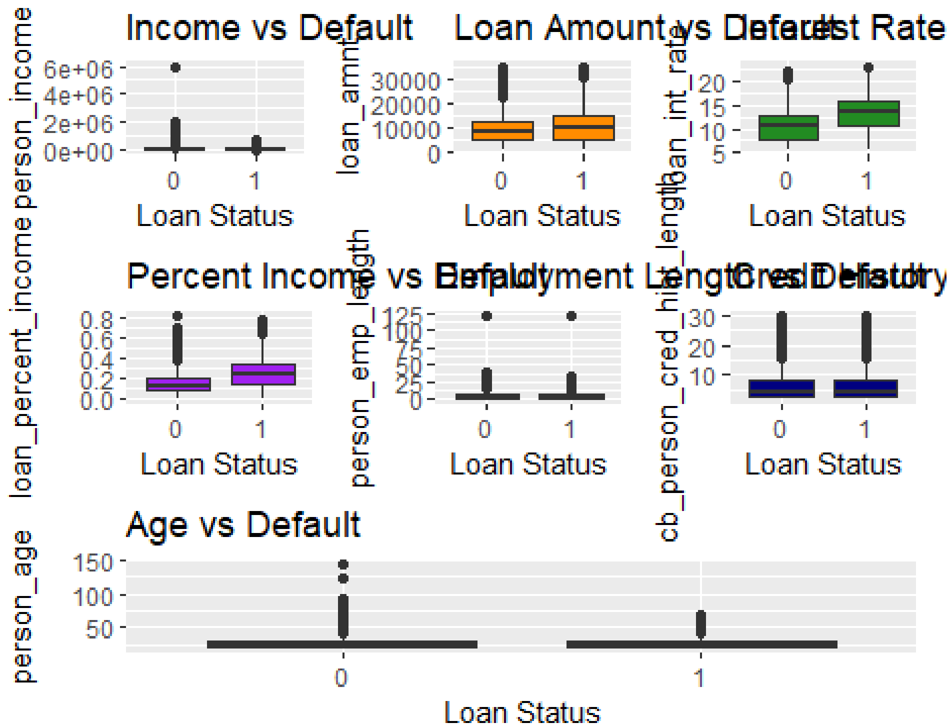
```
...
##  $ person_home_ownership     : chr [1:32581] "RENT" "OWN" "MORTGAGE"
"RENT" ...
##  $ person_emp_length         : num [1:32581] 123 5 1 4 8 2 8 5 8 6 ...
##  $ loan_intent               : chr [1:32581] "PERSONAL" "EDUCATION"
"MEDICAL" "MEDICAL" ...
##  $ loan_grade                : chr [1:32581] "D" "B" "C" "C" ...
##  $ loan_amnt                 : num [1:32581] 35000 1000 5500 35000 35000
2500 35000 35000 35000 1600 ...
##  $ loan_int_rate             : num [1:32581] 16 11.1 12.9 15.2 14.3 ...
##  $ loan_status               : num [1:32581] 1 0 1 1 1 1 1 1 1 1 ...
##  $ loan_percent_income       : num [1:32581] 0.59 0.1 0.57 0.53 0.55 0.25
0.45 0.44 0.42 0.16 ...
##  $ cb_person_default_on_file : chr [1:32581] "Y" "N" "N" "N" ...
##  $ cb_person_cred_hist_length: num [1:32581] 3 2 3 2 4 2 3 4 2 3 ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   person_age = col_double(),
##   ..   person_income = col_double(),
##   ..   person_home_ownership = col_character(),
##   ..   person_emp_length = col_double(),
##   ..   loan_intent = col_character(),
##   ..   loan_grade = col_character(),
##   ..   loan_amnt = col_double(),
##   ..   loan_int_rate = col_double(),
##   ..   loan_status = col_double(),
##   ..   loan_percent_income = col_double(),
##   ..   cb_person_default_on_file = col_character(),
##   ..   cb_person_cred_hist_length = col_double()
##   .. )
##  - attr(*, "problems")=<externalptr>
```

```r
summary <- summary(df)
missing <- colSums(is.na(df))
tv_view <- table(df$loan_status)

p1 <- ggplot(df, aes(x = person_income)) +
  geom_histogram(fill = "steelblue", bins = 30) +
  labs(title = "Person Income")

p2 <- ggplot(df, aes(x = loan_amnt)) +
  geom_histogram(fill = "darkorange", bins = 30) +
  labs(title = "Loan Amount")

p3 <- ggplot(df, aes(x = loan_int_rate)) +
  geom_histogram(fill = "forestgreen", bins = 30) +
  labs(title = "Loan Interest Rate")

p4 <- ggplot(df, aes(x = loan_percent_income)) +
  geom_histogram(fill = "purple", bins = 30) +
```

```
  labs(title = "Loan Percent Income")

p5 <- ggplot(df, aes(x = person_emp_length)) +
  geom_histogram(fill = "firebrick", bins = 30) +
  labs(title = "Employment Length")

p6 <- ggplot(df, aes(x = cb_person_cred_hist_length)) +
  geom_histogram(fill = "navy", bins = 30) +
  labs(title = "Credit History Length")

p7 <- ggplot(df, aes(x = person_age)) +
  geom_histogram(fill = "goldenrod") +
  labs(title = "Age", x = "Age")

(p1 | p2 | p3) /
  (p4 | p5 | p6) /
  (p7)

## Warning: Removed 3116 rows containing non-finite outside the scale range
## (`stat_bin()`).

## Warning: Removed 895 rows containing non-finite outside the scale range
## (`stat_bin()`).

## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

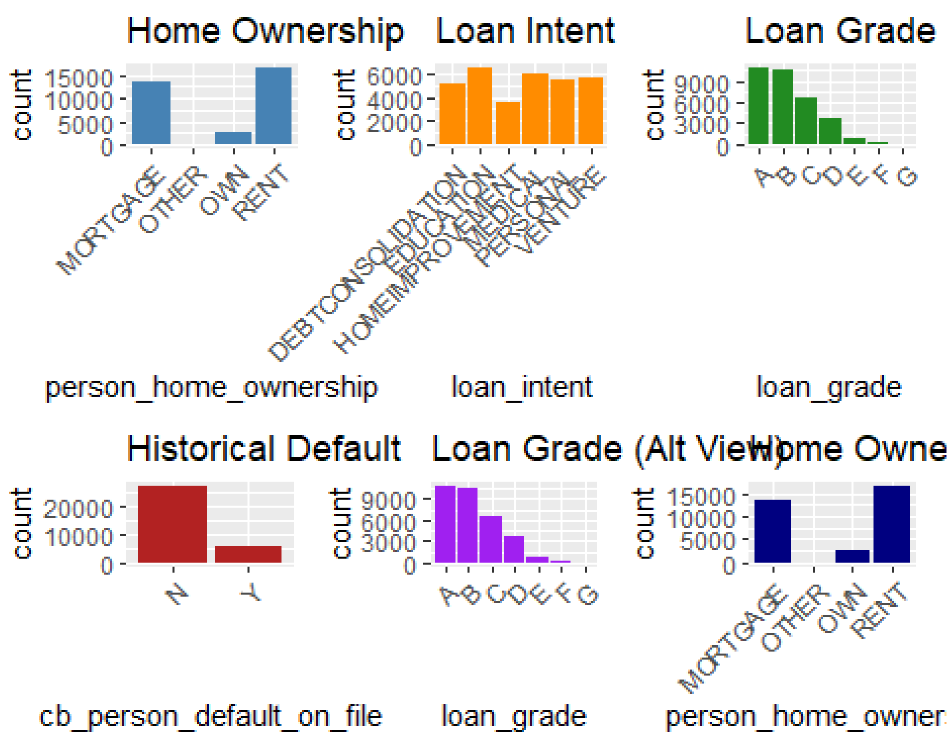


```r
p1 <- ggplot(df, aes(x = factor(loan_status), y = person_income)) +
  geom_boxplot(fill = "steelblue") +
  labs(title = "Income vs Default", x = "Loan Status")

p2 <- ggplot(df, aes(x = factor(loan_status), y = loan_amnt)) +
  geom_boxplot(fill = "darkorange") +
  labs(title = "Loan Amount vs Default", x = "Loan Status")

p3 <- ggplot(df, aes(x = factor(loan_status), y = loan_int_rate)) +
  geom_boxplot(fill = "forestgreen") +
  labs(title = "Interest Rate vs Default", x = "Loan Status")

p4 <- ggplot(df, aes(x = factor(loan_status), y = loan_percent_income)) +
  geom_boxplot(fill = "purple") +
  labs(title = "Percent Income vs Default", x = "Loan Status")

p5 <- ggplot(df, aes(x = factor(loan_status), y = person_emp_length)) +
  geom_boxplot(fill = "firebrick") +
  labs(title = "Employment Length vs Default", x = "Loan Status")

p6 <- ggplot(df, aes(x = factor(loan_status), y =
cb_person_cred_hist_length)) +
  geom_boxplot(fill = "navy") +
  labs(title = "Credit History vs Default", x = "Loan Status")

p7 <- ggplot(df, aes(x = factor(loan_status), y = person_age)) +
  geom_boxplot(fill = "goldenrod") +
  labs(title = "Age vs Default", x = "Loan Status")

(p1 | p2 | p3) /
  (p4 | p5 | p6) /
  (p7)

## Warning: Removed 3116 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 895 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

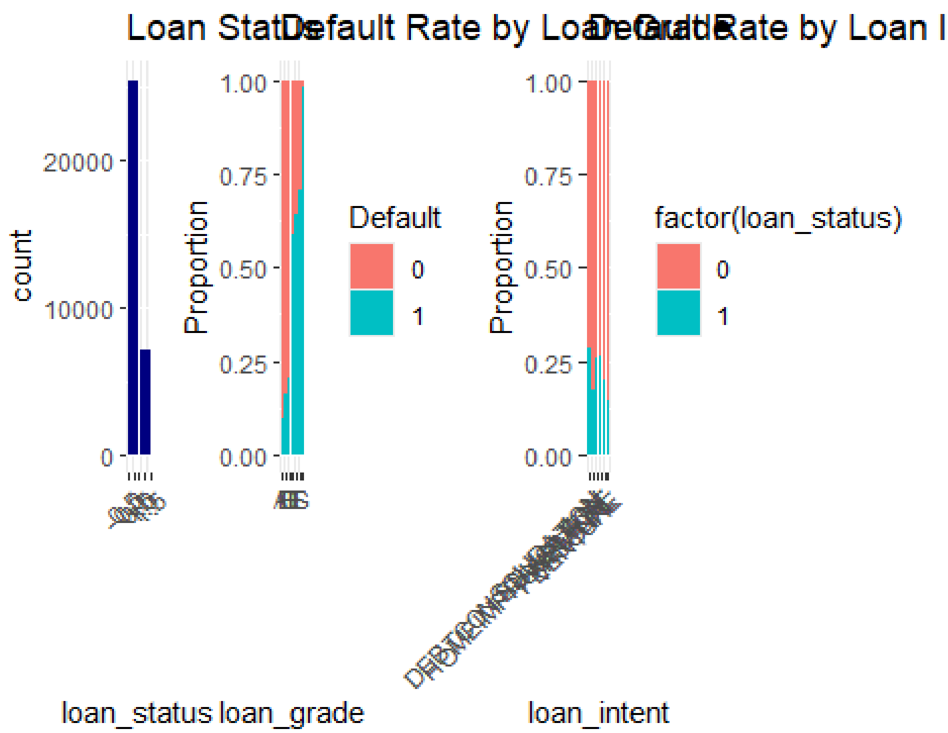



```r
p1 <- ggplot(df, aes(x = person_home_ownership)) +
  geom_bar(fill = "steelblue") +
  labs(title = "Home Ownership") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

p2 <- ggplot(df, aes(x = loan_intent)) +
  geom_bar(fill = "darkorange") +
  labs(title = "Loan Intent") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

p3 <- ggplot(df, aes(x = loan_grade)) +
  geom_bar(fill = "forestgreen") +
  labs(title = "Loan Grade") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

p4 <- ggplot(df, aes(x = cb_person_default_on_file)) +
  geom_bar(fill = "firebrick") +
  labs(title = "Historical Default") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# If you want a 5th and 6th slot filled (optional placeholders or repeats)
p5 <- ggplot(df, aes(x = loan_grade)) +
  geom_bar(fill = "purple") +
  labs(title = "Loan Grade (Alt View)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
p6 <- ggplot(df, aes(x = person_home_ownership)) +
  geom_bar(fill = "navy") +
  labs(title = "Home Ownership (Alt View)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

(p1 | p2 | p3) /
  (p4 | p5 | p6)
```



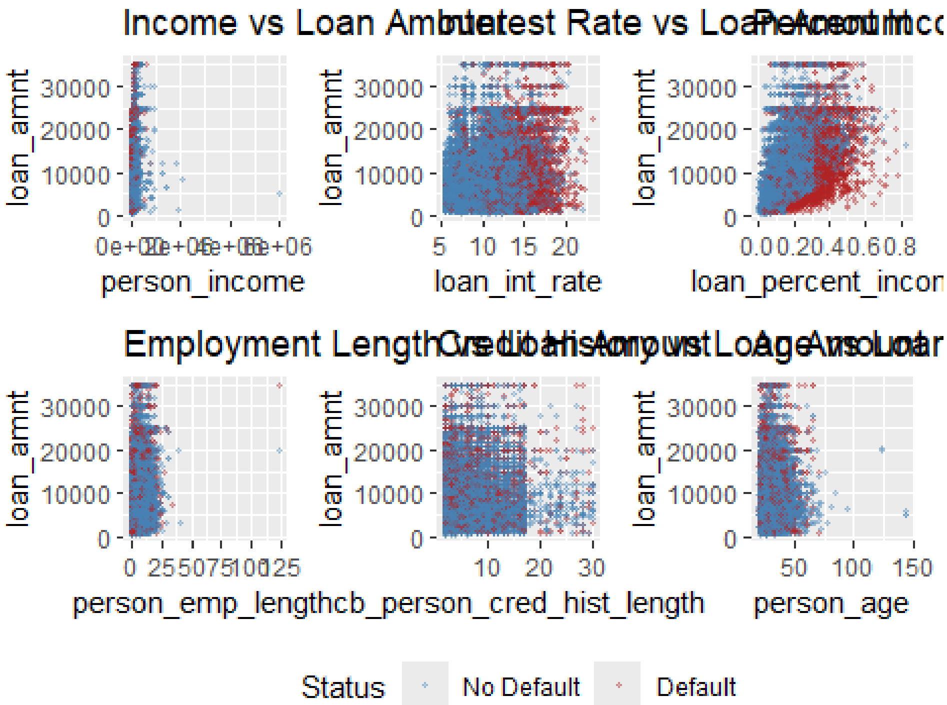

```
p1 <- ggplot(df, aes(x = loan_status)) +
  geom_bar(fill = "navy") +
  labs(title = "Loan Status") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

p2 <- ggplot(df, aes(x = loan_grade, fill = factor(loan_status))) +
  geom_bar(position = "fill") +
  labs(title = "Default Rate by Loan Grade", y = "Proportion", fill =
"Default")

p3 <- ggplot(df, aes(x = loan_intent, fill = factor(loan_status))) +
  geom_bar(position = "fill") +
  labs(title = "Default Rate by Loan Intent", y = "Proportion") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

(p1 | p2 | p3)
```



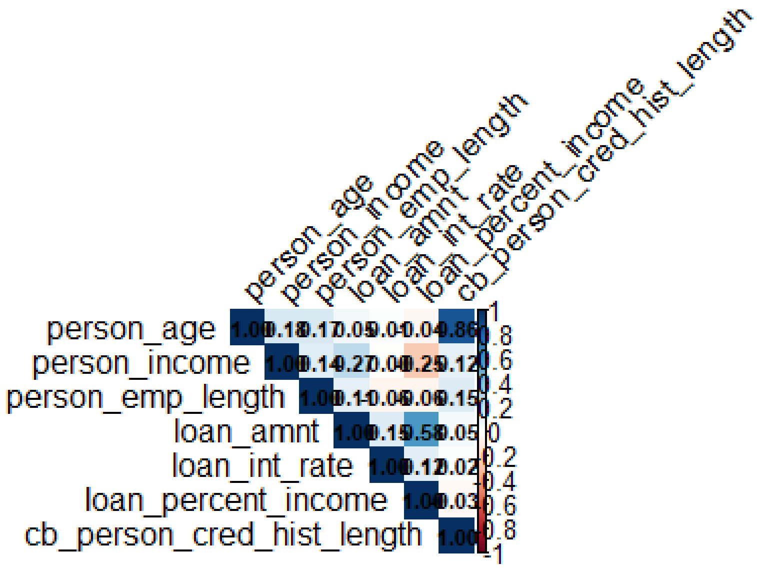

```r
df$loan_status <- factor(df$loan_status, levels = c(0, 1), labels = c("No Default", "Default"))

p1 <- ggplot(df, aes(x = person_income, y = loan_amnt, color = loan_status)) +
  geom_point(alpha = 0.3, size = 0.8) +
  scale_color_manual(values = c("No Default" = "steelblue", "Default" = "firebrick")) +
  labs(title = "Income vs Loan Amount", color = "Status")

p2 <- ggplot(df, aes(x = loan_int_rate, y = loan_amnt, color = loan_status)) +
  geom_point(alpha = 0.3, size = 0.8) +
  scale_color_manual(values = c("No Default" = "steelblue", "Default" = "firebrick")) +
  labs(title = "Interest Rate vs Loan Amount", color = "Status")

p3 <- ggplot(df, aes(x = loan_percent_income, y = loan_amnt, color = loan_status)) +
  geom_point(alpha = 0.3, size = 0.8) +
  scale_color_manual(values = c("No Default" = "steelblue", "Default" = "firebrick")) +
  labs(title = "Percent Income vs Loan Amount", color = "Status")

p4 <- ggplot(df, aes(x = person_emp_length, y = loan_amnt, color = loan_status)) +
```

```r
  geom_point(alpha = 0.3, size = 0.8) +
  scale_color_manual(values = c("No Default" = "steelblue", "Default" =
"firebrick")) +
  labs(title = "Employment Length vs Loan Amount", color = "Status")

p5 <- ggplot(df, aes(x = cb_person_cred_hist_length, y = loan_amnt, color =
loan_status)) +
  geom_point(alpha = 0.3, size = 0.8) +
  scale_color_manual(values = c("No Default" = "steelblue", "Default" =
"firebrick")) +
  labs(title = "Credit History vs Loan Amount", color = "Status")

p6 <- ggplot(df, aes(x = person_age, y = loan_amnt, color = loan_status)) +
  geom_point(alpha = 0.3, size = 0.8) +
  scale_color_manual(values = c("No Default" = "steelblue", "Default" =
"firebrick")) +
  labs(title = "Age vs Loan Amount", color = "Status")

(p1 | p2 | p3) /
  (p4 | p5 | p6) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```

```
## Warning: Removed 3116 rows containing missing values or values outside the
scale range
## (`geom_point()`).

## Warning: Removed 895 rows containing missing values or values outside the
scale range
## (`geom_point()`).
```

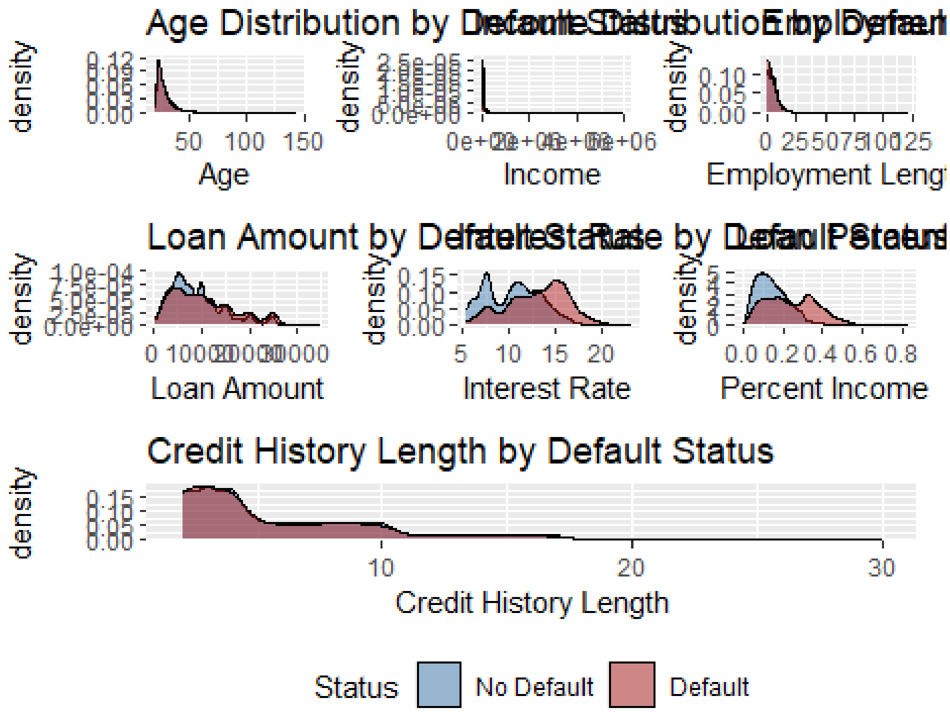



```
corr_matrix <- cor(df[, sapply(df, is.numeric)], use = "complete.obs")

corrplot(corr_matrix,
         method = "color",
         type = "upper",
         tl.col = "black",
         tl.srt = 45,
         addCoef.col = "black",    # show values
         number.cex = 0.7)         # size of numbers
```

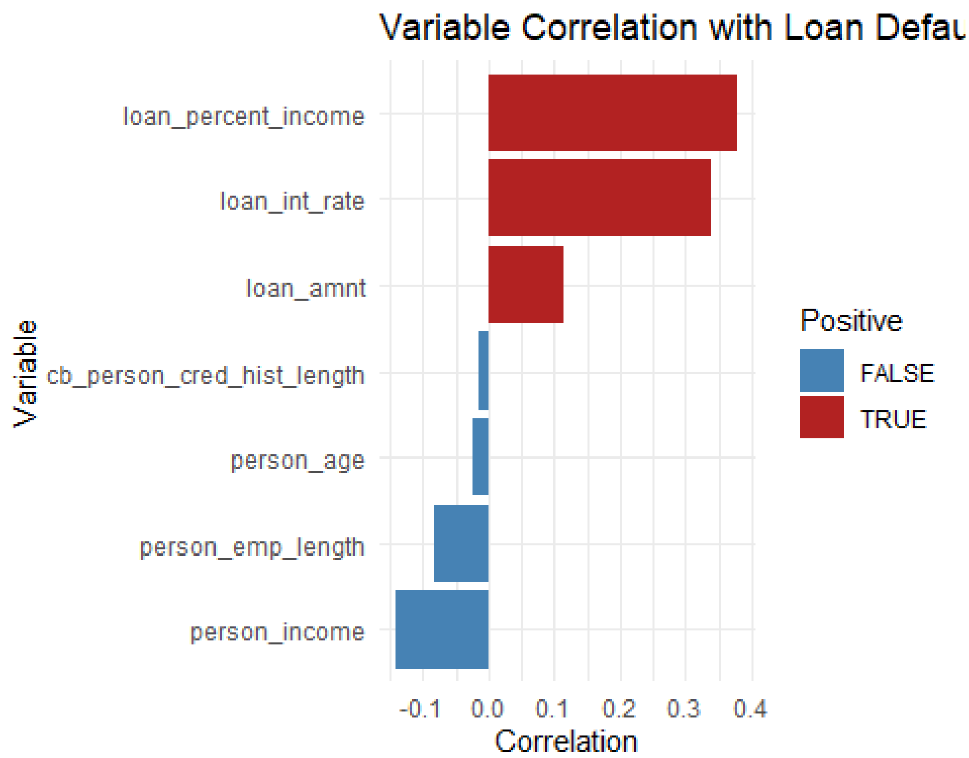



```
p1 <- ggplot(df, aes(x = person_age, fill = loan_status)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("No Default" = "steelblue", "Default" =
"firebrick")) +
  labs(title = "Age Distribution by Default Status", x = "Age", fill =
"Status")

p2 <- ggplot(df, aes(x = person_income, fill = loan_status)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("No Default" = "steelblue", "Default" =
"firebrick")) +
  labs(title = "Income Distribution by Default Status", x = "Income", fill =
"Status")

p3 <- ggplot(df, aes(x = person_emp_length, fill = loan_status)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("No Default" = "steelblue", "Default" =
"firebrick")) +
  labs(title = "Employment Length by Default Status", x = "Employment
Length", fill = "Status")

p4 <- ggplot(df, aes(x = loan_amnt, fill = loan_status)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("No Default" = "steelblue", "Default" =
"firebrick")) +
  labs(title = "Loan Amount by Default Status", x = "Loan Amount", fill =
```

```
"Status")

p5 <- ggplot(df, aes(x = loan_int_rate, fill = loan_status)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("No Default" = "steelblue", "Default" =
"firebrick")) +
  labs(title = "Interest Rate by Default Status", x = "Interest Rate", fill =
"Status")

p6 <- ggplot(df, aes(x = loan_percent_income, fill = loan_status)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("No Default" = "steelblue", "Default" =
"firebrick")) +
  labs(title = "Loan Percent Income by Default Status", x = "Percent Income",
fill = "Status")

p7 <- ggplot(df, aes(x = cb_person_cred_hist_length, fill = loan_status)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("No Default" = "steelblue", "Default" =
"firebrick")) +
  labs(title = "Credit History Length by Default Status", x = "Credit History
Length", fill = "Status")

(p1 | p2 | p3) /
  (p4 | p5 | p6) /
  (p7) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")

## Warning: Removed 895 rows containing non-finite outside the scale range
## (`stat_density()`).

## Warning: Removed 3116 rows containing non-finite outside the scale range
## (`stat_density()`).
```




```r
num_df <- df[, sapply(df, is.numeric)]
num_df$loan_status <- as.integer(df$loan_status == "Default")

cors <- cor(num_df, use = "complete.obs")[, "loan_status"]
cors <- sort(cors[names(cors) != "loan_status"], decreasing = TRUE)
cors_df <- data.frame(variable = names(cors), correlation = cors)

ggplot(cors_df, aes(x = reorder(variable, correlation), y = correlation, fill
= correlation > 0)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("TRUE" = "firebrick", "FALSE" = "steelblue"))
+
  coord_flip() +
  labs(title = "Variable Correlation with Loan Default",
       x = "Variable", y = "Correlation", fill = "Positive") +
  theme_minimal()
```




```r
cap_outliers <- function(x) {
  q1 <- quantile(x, 0.25, na.rm = TRUE)
  q3 <- quantile(x, 0.75, na.rm = TRUE)
  iqr <- q3 - q1
  pmin(pmax(x, q1 - 1.5 * iqr), q3 + 1.5 * iqr)
}

numeric_cols <- names(df)[sapply(df, is.numeric)]
df[numeric_cols] <- lapply(df[numeric_cols], cap_outliers)

miss_summary <- colSums(is.na(df))
miss_pct <- round(miss_summary / nrow(df) * 100, 2)
print(miss_pct)
```

```
##                 person_age              person_income
##                       0.00                       0.00
##      person_home_ownership          person_emp_length
##                       0.00                       2.75
##                loan_intent                 loan_grade
##                       0.00                       0.00
##                  loan_amnt              loan_int_rate
##                       0.00                       9.56
##                loan_status        loan_percent_income
##                       0.00                       0.00
##  cb_person_default_on_file cb_person_cred_hist_length
##                       0.00                       0.00
```

```r
grade_map <- c("A" = 1, "B" = 2, "C" = 3, "D" = 4, "E" = 5, "F" = 6, "G" = 7)
df$loan_grade_encoded <- grade_map[df$loan_grade]

df$cb_person_default_on_file_encoded <- ifelse(df$cb_person_default_on_file
== "Y", 1, 0)

df$home_MORTGAGE <- ifelse(df$person_home_ownership == "MORTGAGE", 1, 0)
df$home_OWN      <- ifelse(df$person_home_ownership == "OWN", 1, 0)
df$home_OTHER    <- ifelse(df$person_home_ownership == "OTHER", 1, 0)

df$intent_DEBTCONSOLIDATION <- ifelse(df$loan_intent == "DEBTCONSOLIDATION",
1, 0)
df$intent_HOMEIMPROVEMENT   <- ifelse(df$loan_intent == "HOMEIMPROVEMENT", 1,
0)
df$intent_MEDICAL           <- ifelse(df$loan_intent == "MEDICAL", 1, 0)
df$intent_PERSONAL          <- ifelse(df$loan_intent == "PERSONAL", 1, 0)
df$intent_VENTURE           <- ifelse(df$loan_intent == "VENTURE", 1, 0)

df <- df[, !sapply(df, is.character)]

df$person_emp_length[is.na(df$person_emp_length)] <-
median(df$person_emp_length, na.rm = TRUE)
df$loan_int_rate[is.na(df$loan_int_rate)]         <- median(df$loan_int_rate,
na.rm = TRUE)

df$loan_status <- ifelse(df$loan_status == "Default", 1, 0)

set.seed(123)
train_idx <- sample(1:nrow(df), size = 0.8 * nrow(df))
train <- df[train_idx, ]
test  <- df[-train_idx, ]

train_x <- as.matrix(train[, !names(train) %in% "loan_status"])
train_y <- train$loan_status

test_x  <- as.matrix(test[, !names(test) %in% "loan_status"])
test_y  <- test$loan_status

dtrain  <- xgb.DMatrix(data = train_x, label = train_y)
dtest   <- xgb.DMatrix(data = test_x,  label = test_y)

params <- list(
  objective        = "binary:logistic",
  eval_metric      = "auc",
  eta              = 0.1,
  max_depth        = 6,
  subsample        = 0.8,
  colsample_bytree = 0.8,
  min_child_weight = 5,
```

```r
  gamma             = 1
)

evals <- list(train = dtrain, eval = dtest)

xgb_model <- xgb.train(
  params                = params,
  data                  = dtrain,
  nrounds               = 500,
  evals                 = evals,
  early_stopping_rounds = 30,
  verbose               = 1
)
```

```
## Multiple eval metrics are present. Will use eval_auc for early stopping.
## Will train until eval_auc hasn't improved in 30 rounds.
##
## [1]  train-auc:0.888651  eval-auc:0.888497
## [2]  train-auc:0.896976  eval-auc:0.896106
## [3]  train-auc:0.907167  eval-auc:0.907983
## [4]  train-auc:0.907662  eval-auc:0.909533
## [5]  train-auc:0.912757  eval-auc:0.913057
## [6]  train-auc:0.915756  eval-auc:0.916125
## [7]  train-auc:0.917738  eval-auc:0.918591
## [8]  train-auc:0.918098  eval-auc:0.919005
## [9]  train-auc:0.919905  eval-auc:0.920888
## [10] train-auc:0.921072  eval-auc:0.922683
## [11] train-auc:0.922431  eval-auc:0.923311
## [12] train-auc:0.922280  eval-auc:0.923426
## [13] train-auc:0.922564  eval-auc:0.923793
## [14] train-auc:0.923618  eval-auc:0.924736
## [15] train-auc:0.924016  eval-auc:0.924723
## [16] train-auc:0.924893  eval-auc:0.925741
## [17] train-auc:0.925768  eval-auc:0.927007
## [18] train-auc:0.925607  eval-auc:0.926857
## [19] train-auc:0.926667  eval-auc:0.928001
## [20] train-auc:0.927255  eval-auc:0.928270
## [21] train-auc:0.927972  eval-auc:0.928955
## [22] train-auc:0.928174  eval-auc:0.929362
## [23] train-auc:0.928505  eval-auc:0.929743
## [24] train-auc:0.929343  eval-auc:0.930474
## [25] train-auc:0.930263  eval-auc:0.930813
## [26] train-auc:0.930992  eval-auc:0.931139
## [27] train-auc:0.931671  eval-auc:0.931819
## [28] train-auc:0.932169  eval-auc:0.932218
## [29] train-auc:0.932532  eval-auc:0.932035
## [30] train-auc:0.933429  eval-auc:0.932520
## [31] train-auc:0.933793  eval-auc:0.933023
## [32] train-auc:0.934733  eval-auc:0.933966
## [33] train-auc:0.935294  eval-auc:0.934187
```

```
## [34] train-auc:0.935666  eval-auc:0.934541
## [35] train-auc:0.936658  eval-auc:0.935184
## [36] train-auc:0.937063  eval-auc:0.935473
## [37] train-auc:0.937485  eval-auc:0.935848
## [38] train-auc:0.939239  eval-auc:0.937581
## [39] train-auc:0.939620  eval-auc:0.937790
## [40] train-auc:0.939904  eval-auc:0.937769
## [41] train-auc:0.940291  eval-auc:0.938262
## [42] train-auc:0.940650  eval-auc:0.938484
## [43] train-auc:0.940924  eval-auc:0.938615
## [44] train-auc:0.941302  eval-auc:0.938706
## [45] train-auc:0.942390  eval-auc:0.939691
## [46] train-auc:0.943420  eval-auc:0.940382
## [47] train-auc:0.943669  eval-auc:0.940522
## [48] train-auc:0.944107  eval-auc:0.940653
## [49] train-auc:0.944561  eval-auc:0.940788
## [50] train-auc:0.944749  eval-auc:0.940855
## [51] train-auc:0.945166  eval-auc:0.940875
## [52] train-auc:0.945528  eval-auc:0.940997
## [53] train-auc:0.945806  eval-auc:0.941123
## [54] train-auc:0.945997  eval-auc:0.941224
## [55] train-auc:0.946209  eval-auc:0.941241
## [56] train-auc:0.947114  eval-auc:0.942163
## [57] train-auc:0.947343  eval-auc:0.942058
## [58] train-auc:0.947882  eval-auc:0.942455
## [59] train-auc:0.948287  eval-auc:0.942418
## [60] train-auc:0.948638  eval-auc:0.942638
## [61] train-auc:0.949178  eval-auc:0.942614
## [62] train-auc:0.949369  eval-auc:0.942746
## [63] train-auc:0.950386  eval-auc:0.943370
## [64] train-auc:0.950759  eval-auc:0.943521
## [65] train-auc:0.950912  eval-auc:0.943582
## [66] train-auc:0.951185  eval-auc:0.943627
## [67] train-auc:0.951760  eval-auc:0.943600
## [68] train-auc:0.952132  eval-auc:0.943752
## [69] train-auc:0.952212  eval-auc:0.943759
## [70] train-auc:0.952356  eval-auc:0.943805
## [71] train-auc:0.952615  eval-auc:0.943863
## [72] train-auc:0.953125  eval-auc:0.943885
## [73] train-auc:0.953460  eval-auc:0.944178
## [74] train-auc:0.953686  eval-auc:0.944202
## [75] train-auc:0.953919  eval-auc:0.944068
## [76] train-auc:0.954339  eval-auc:0.943882
## [77] train-auc:0.954557  eval-auc:0.943851
## [78] train-auc:0.954954  eval-auc:0.944181
## [79] train-auc:0.955175  eval-auc:0.944163
## [80] train-auc:0.955246  eval-auc:0.944148
## [81] train-auc:0.955641  eval-auc:0.944150
## [82] train-auc:0.956213  eval-auc:0.944388
## [83] train-auc:0.956459  eval-auc:0.944283
```

```
## [84] train-auc:0.956959  eval-auc:0.944930
## [85] train-auc:0.957267  eval-auc:0.945175
## [86] train-auc:0.957418  eval-auc:0.945188
## [87] train-auc:0.957591  eval-auc:0.945083
## [88] train-auc:0.957672  eval-auc:0.945138
## [89] train-auc:0.958262  eval-auc:0.945349
## [90] train-auc:0.958334  eval-auc:0.945409
## [91] train-auc:0.958616  eval-auc:0.945622
## [92] train-auc:0.958895  eval-auc:0.945730
## [93] train-auc:0.959213  eval-auc:0.945747
## [94] train-auc:0.959628  eval-auc:0.945909
## [95] train-auc:0.959938  eval-auc:0.946070
## [96] train-auc:0.960279  eval-auc:0.946269
## [97] train-auc:0.960520  eval-auc:0.946388
## [98] train-auc:0.960618  eval-auc:0.946367
## [99] train-auc:0.960983  eval-auc:0.946516
## [100]    train-auc:0.961262  eval-auc:0.946690
## [101]    train-auc:0.961463  eval-auc:0.946700
## [102]    train-auc:0.961598  eval-auc:0.946666
## [103]    train-auc:0.962095  eval-auc:0.946690
## [104]    train-auc:0.962368  eval-auc:0.946842
## [105]    train-auc:0.962420  eval-auc:0.946823
## [106]    train-auc:0.962783  eval-auc:0.946778
## [107]    train-auc:0.963100  eval-auc:0.946809
## [108]    train-auc:0.963122  eval-auc:0.946763
## [109]    train-auc:0.963538  eval-auc:0.946918
## [110]    train-auc:0.963877  eval-auc:0.947191
## [111]    train-auc:0.964128  eval-auc:0.947224
## [112]    train-auc:0.964325  eval-auc:0.947205
## [113]    train-auc:0.964786  eval-auc:0.947331
## [114]    train-auc:0.965061  eval-auc:0.947238
## [115]    train-auc:0.965350  eval-auc:0.947481
## [116]    train-auc:0.965483  eval-auc:0.947440
## [117]    train-auc:0.965922  eval-auc:0.947420
## [118]    train-auc:0.966110  eval-auc:0.947473
## [119]    train-auc:0.966250  eval-auc:0.947705
## [120]    train-auc:0.966255  eval-auc:0.947721
## [121]    train-auc:0.966526  eval-auc:0.947781
## [122]    train-auc:0.966720  eval-auc:0.947837
## [123]    train-auc:0.966804  eval-auc:0.947913
## [124]    train-auc:0.966951  eval-auc:0.947969
## [125]    train-auc:0.967115  eval-auc:0.947976
## [126]    train-auc:0.967269  eval-auc:0.948237
## [127]    train-auc:0.967467  eval-auc:0.948178
## [128]    train-auc:0.967670  eval-auc:0.948095
## [129]    train-auc:0.967783  eval-auc:0.948023
## [130]    train-auc:0.967945  eval-auc:0.948023
## [131]    train-auc:0.968359  eval-auc:0.948027
## [132]    train-auc:0.968582  eval-auc:0.948026
## [133]    train-auc:0.968797  eval-auc:0.947949
```

```
## [134]	train-auc:0.968919	eval-auc:0.947976
## [135]	train-auc:0.968998	eval-auc:0.948061
## [136]	train-auc:0.969217	eval-auc:0.948038
## [137]	train-auc:0.969269	eval-auc:0.948124
## [138]	train-auc:0.969464	eval-auc:0.948011
## [139]	train-auc:0.969560	eval-auc:0.948087
## [140]	train-auc:0.969785	eval-auc:0.948066
## [141]	train-auc:0.970048	eval-auc:0.948094
## [142]	train-auc:0.970249	eval-auc:0.948173
## [143]	train-auc:0.970325	eval-auc:0.948237
## [144]	train-auc:0.970635	eval-auc:0.948154
## [145]	train-auc:0.970764	eval-auc:0.947979
## [146]	train-auc:0.970958	eval-auc:0.947926
## [147]	train-auc:0.971143	eval-auc:0.947883
## [148]	train-auc:0.971294	eval-auc:0.948095
## [149]	train-auc:0.971546	eval-auc:0.948161
## [150]	train-auc:0.971708	eval-auc:0.948102
## [151]	train-auc:0.971839	eval-auc:0.948039
## [152]	train-auc:0.971895	eval-auc:0.948016
## [153]	train-auc:0.972020	eval-auc:0.948083
## [154]	train-auc:0.972137	eval-auc:0.948035
## [155]	train-auc:0.972304	eval-auc:0.948051
## [156]	train-auc:0.972512	eval-auc:0.948133
## [157]	train-auc:0.972717	eval-auc:0.948128
## [158]	train-auc:0.972970	eval-auc:0.948092
## [159]	train-auc:0.973164	eval-auc:0.947996
## [160]	train-auc:0.973251	eval-auc:0.948037
## [161]	train-auc:0.973323	eval-auc:0.948014
## [162]	train-auc:0.973374	eval-auc:0.948115
## [163]	train-auc:0.973527	eval-auc:0.948238
## [164]	train-auc:0.973628	eval-auc:0.948357
## [165]	train-auc:0.973850	eval-auc:0.948396
## [166]	train-auc:0.973908	eval-auc:0.948419
## [167]	train-auc:0.974094	eval-auc:0.948434
## [168]	train-auc:0.974325	eval-auc:0.948491
## [169]	train-auc:0.974404	eval-auc:0.948520
## [170]	train-auc:0.974548	eval-auc:0.948545
## [171]	train-auc:0.974721	eval-auc:0.948407
## [172]	train-auc:0.974840	eval-auc:0.948336
## [173]	train-auc:0.974936	eval-auc:0.948307
## [174]	train-auc:0.975020	eval-auc:0.948369
## [175]	train-auc:0.975298	eval-auc:0.948482
## [176]	train-auc:0.975450	eval-auc:0.948418
## [177]	train-auc:0.975690	eval-auc:0.948407
## [178]	train-auc:0.975849	eval-auc:0.948269
## [179]	train-auc:0.975972	eval-auc:0.948230
## [180]	train-auc:0.976111	eval-auc:0.948165
## [181]	train-auc:0.976266	eval-auc:0.948203
## [182]	train-auc:0.976488	eval-auc:0.948347
## [183]	train-auc:0.976488	eval-auc:0.948347
```

```
## [184]	train-auc:0.976511	eval-auc:0.948316
## [185]	train-auc:0.976571	eval-auc:0.948429
## [186]	train-auc:0.976728	eval-auc:0.948416
## [187]	train-auc:0.976840	eval-auc:0.948555
## [188]	train-auc:0.976897	eval-auc:0.948465
## [189]	train-auc:0.977040	eval-auc:0.948470
## [190]	train-auc:0.977211	eval-auc:0.948446
## [191]	train-auc:0.977204	eval-auc:0.948439
## [192]	train-auc:0.977204	eval-auc:0.948439
## [193]	train-auc:0.977260	eval-auc:0.948532
## [194]	train-auc:0.977266	eval-auc:0.948541
## [195]	train-auc:0.977391	eval-auc:0.948595
## [196]	train-auc:0.977534	eval-auc:0.948632
## [197]	train-auc:0.977703	eval-auc:0.948651
## [198]	train-auc:0.977749	eval-auc:0.948730
## [199]	train-auc:0.977896	eval-auc:0.948649
## [200]	train-auc:0.977896	eval-auc:0.948649
## [201]	train-auc:0.978047	eval-auc:0.948678
## [202]	train-auc:0.978051	eval-auc:0.948681
## [203]	train-auc:0.978057	eval-auc:0.948727
## [204]	train-auc:0.978138	eval-auc:0.948780
## [205]	train-auc:0.978223	eval-auc:0.948822
## [206]	train-auc:0.978380	eval-auc:0.948859
## [207]	train-auc:0.978384	eval-auc:0.948872
## [208]	train-auc:0.978549	eval-auc:0.948889
## [209]	train-auc:0.978547	eval-auc:0.948909
## [210]	train-auc:0.978592	eval-auc:0.948885
## [211]	train-auc:0.978705	eval-auc:0.948869
## [212]	train-auc:0.978871	eval-auc:0.948846
## [213]	train-auc:0.978871	eval-auc:0.948846
## [214]	train-auc:0.978912	eval-auc:0.948881
## [215]	train-auc:0.979061	eval-auc:0.948780
## [216]	train-auc:0.979098	eval-auc:0.948880
## [217]	train-auc:0.979275	eval-auc:0.948739
## [218]	train-auc:0.979393	eval-auc:0.948674
## [219]	train-auc:0.979446	eval-auc:0.948682
## [220]	train-auc:0.979590	eval-auc:0.948794
## [221]	train-auc:0.979680	eval-auc:0.948846
## [222]	train-auc:0.979737	eval-auc:0.948903
## [223]	train-auc:0.979787	eval-auc:0.949021
## [224]	train-auc:0.979787	eval-auc:0.949021
## [225]	train-auc:0.979879	eval-auc:0.949026
## [226]	train-auc:0.979930	eval-auc:0.949041
## [227]	train-auc:0.980021	eval-auc:0.948939
## [228]	train-auc:0.980192	eval-auc:0.949020
## [229]	train-auc:0.980192	eval-auc:0.949020
## [230]	train-auc:0.980284	eval-auc:0.949046
## [231]	train-auc:0.980394	eval-auc:0.949033
## [232]	train-auc:0.980549	eval-auc:0.949092
## [233]	train-auc:0.980735	eval-auc:0.949073
```

```
## [234]	train-auc:0.980765	eval-auc:0.949144
## [235]	train-auc:0.980901	eval-auc:0.949032
## [236]	train-auc:0.980901	eval-auc:0.949032
## [237]	train-auc:0.980959	eval-auc:0.949060
## [238]	train-auc:0.980959	eval-auc:0.949060
## [239]	train-auc:0.981080	eval-auc:0.949005
## [240]	train-auc:0.981087	eval-auc:0.948991
## [241]	train-auc:0.981183	eval-auc:0.949075
## [242]	train-auc:0.981183	eval-auc:0.949075
## [243]	train-auc:0.981183	eval-auc:0.949075
## [244]	train-auc:0.981273	eval-auc:0.949156
## [245]	train-auc:0.981273	eval-auc:0.949156
## [246]	train-auc:0.981273	eval-auc:0.949156
## [247]	train-auc:0.981273	eval-auc:0.949156
## [248]	train-auc:0.981273	eval-auc:0.949156
## [249]	train-auc:0.981273	eval-auc:0.949156
## [250]	train-auc:0.981327	eval-auc:0.949116
## [251]	train-auc:0.981514	eval-auc:0.949200
## [252]	train-auc:0.981706	eval-auc:0.949124
## [253]	train-auc:0.981793	eval-auc:0.949172
## [254]	train-auc:0.981842	eval-auc:0.949229
## [255]	train-auc:0.981900	eval-auc:0.949254
## [256]	train-auc:0.982006	eval-auc:0.949275
## [257]	train-auc:0.982036	eval-auc:0.949290
## [258]	train-auc:0.982086	eval-auc:0.949307
## [259]	train-auc:0.982268	eval-auc:0.949304
## [260]	train-auc:0.982342	eval-auc:0.949350
## [261]	train-auc:0.982421	eval-auc:0.949270
## [262]	train-auc:0.982462	eval-auc:0.949225
## [263]	train-auc:0.982514	eval-auc:0.949242
## [264]	train-auc:0.982509	eval-auc:0.949241
## [265]	train-auc:0.982695	eval-auc:0.949154
## [266]	train-auc:0.982691	eval-auc:0.949152
## [267]	train-auc:0.982730	eval-auc:0.949135
## [268]	train-auc:0.982799	eval-auc:0.949149
## [269]	train-auc:0.982907	eval-auc:0.949047
## [270]	train-auc:0.982907	eval-auc:0.949047
## [271]	train-auc:0.982912	eval-auc:0.949060
## [272]	train-auc:0.982910	eval-auc:0.949067
## [273]	train-auc:0.982940	eval-auc:0.949077
## [274]	train-auc:0.982940	eval-auc:0.949077
## [275]	train-auc:0.982940	eval-auc:0.949077
## [276]	train-auc:0.982980	eval-auc:0.949120
## [277]	train-auc:0.983093	eval-auc:0.949118
## [278]	train-auc:0.983130	eval-auc:0.949139
## [279]	train-auc:0.983156	eval-auc:0.949117
## [280]	train-auc:0.983266	eval-auc:0.949091
## [281]	train-auc:0.983266	eval-auc:0.949091
## [282]	train-auc:0.983295	eval-auc:0.949153
## [283]	train-auc:0.983288	eval-auc:0.949119
```

```
## [284]    train-auc:0.983343  eval-auc:0.949187
## [285]    train-auc:0.983343  eval-auc:0.949187
## [286]    train-auc:0.983399  eval-auc:0.949184
## [287]    train-auc:0.983548  eval-auc:0.949205
## [288]    train-auc:0.983595  eval-auc:0.949220
## [289]    train-auc:0.983660  eval-auc:0.949235
## Stopping. Best iteration:
## [290]    train-auc:0.983698  eval-auc:0.949211
##
## [290]    train-auc:0.983698  eval-auc:0.949211
```

```
pred_prob  <- predict(xgb_model, dtest)
pred_class <- ifelse(pred_prob > 0.5, 1, 0)

confusionMatrix(factor(pred_class), factor(test_y), positive = "1")
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##          0 5022  365
##          1   53 1077
##
##                Accuracy : 0.9359
##                  95% CI : (0.9296, 0.9417)
##     No Information Rate : 0.7787
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.7983
##
##  Mcnemar's Test P-Value : < 2.2e-16
##
##             Sensitivity : 0.7469
##             Specificity : 0.9896
##          Pos Pred Value : 0.9531
##          Neg Pred Value : 0.9322
##              Prevalence : 0.2213
##          Detection Rate : 0.1653
##    Detection Prevalence : 0.1734
##       Balanced Accuracy : 0.8682
##
##        'Positive' Class : 1
##
```

```
roc_obj <- roc(test_y, pred_prob)
```

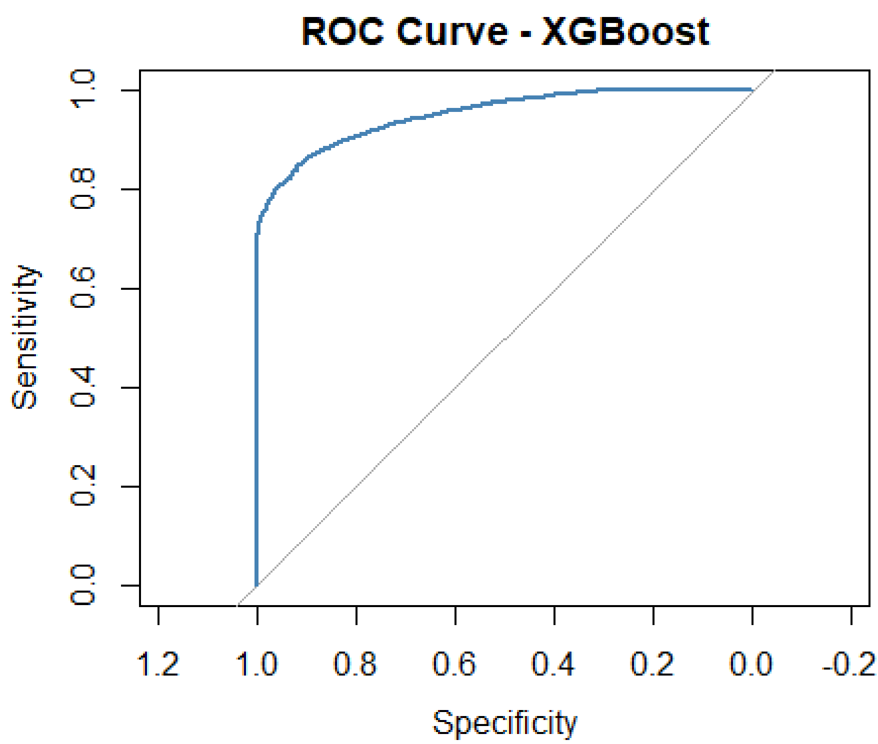
```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

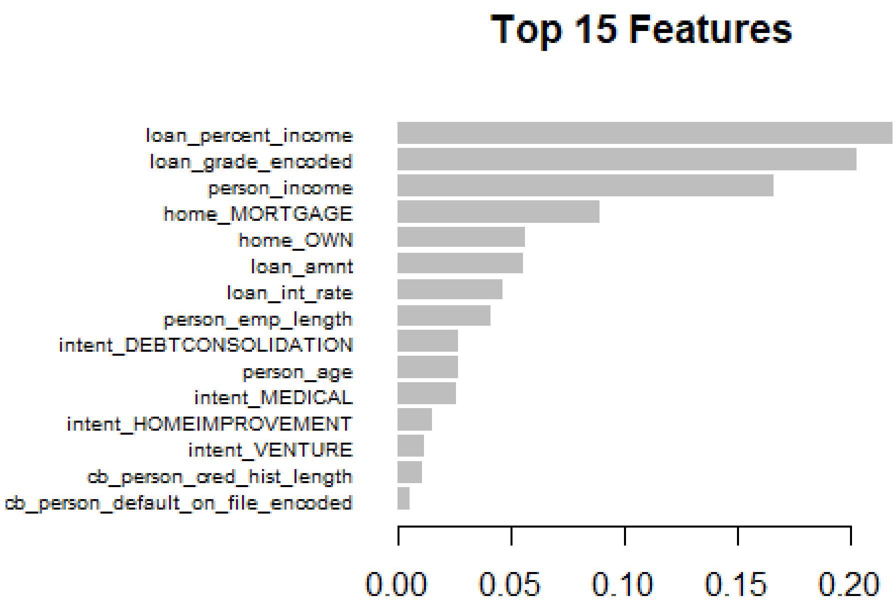
```
cat("AUC:", auc(roc_obj), "\n")
```

```
## AUC: 0.9493498
```

```r
plot(roc_obj, col = "steelblue", main = "ROC Curve - XGBoost")
```




```r
importance_matrix <- xgb.importance(model = xgb_model)
xgb.plot.importance(importance_matrix, top_n = 15, main = "Top 15 Features")
```




```r
pred_prob  <- predict(xgb_model, dtest)
pred_class <- ifelse(pred_prob > 0.3, 1, 0)

confusionMatrix(factor(pred_class), factor(test_y), positive = "1")

## Confusion Matrix and Statistics
##
##             Reference
## Prediction    0    1
##            0 4888  297
##            1  187 1145
##
##                Accuracy : 0.9257
##                  95% CI : (0.9191, 0.932)
##     No Information Rate : 0.7787
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.7784
##
##  Mcnemar's Test P-Value : 7.25e-07
##
##             Sensitivity : 0.7940
##             Specificity : 0.9632
##          Pos Pred Value : 0.8596
##          Neg Pred Value : 0.9427
##              Prevalence : 0.2213
##          Detection Rate : 0.1757
```

```
##    Detection Prevalence : 0.2044
##       Balanced Accuracy : 0.8786
##
##        'Positive' Class : 1
##
```

```r
roc_obj <- roc(test_y, pred_prob)
```

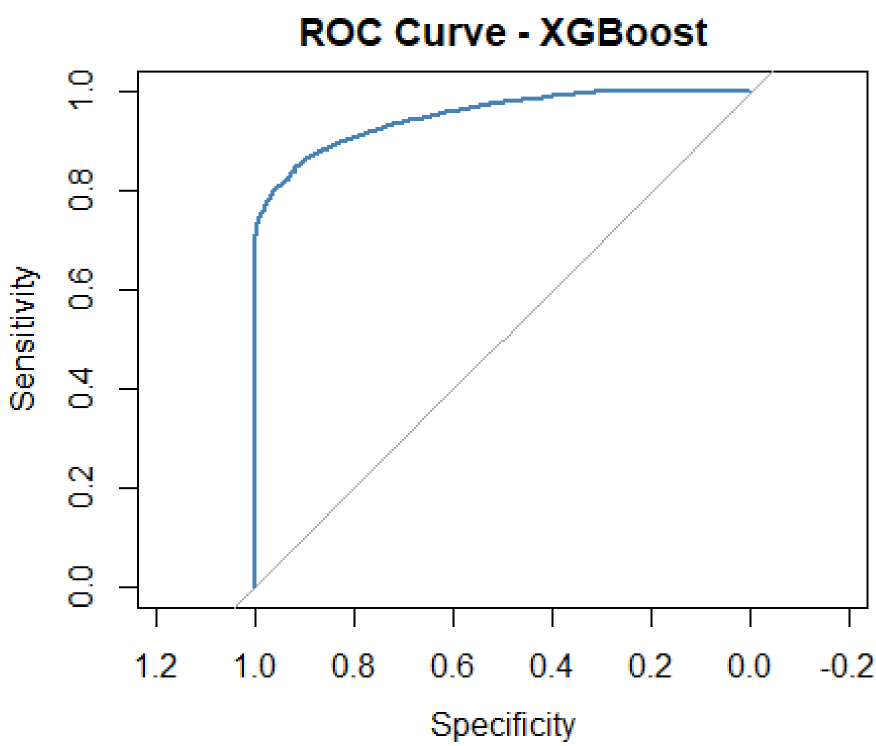
```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

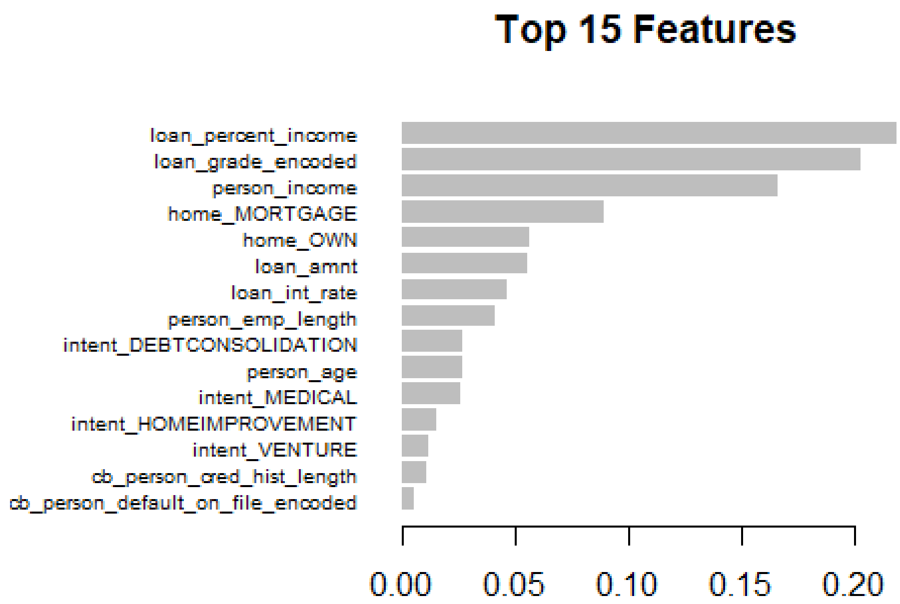
```r
cat("AUC:", auc(roc_obj), "\n")
```

```
## AUC: 0.9493498
```

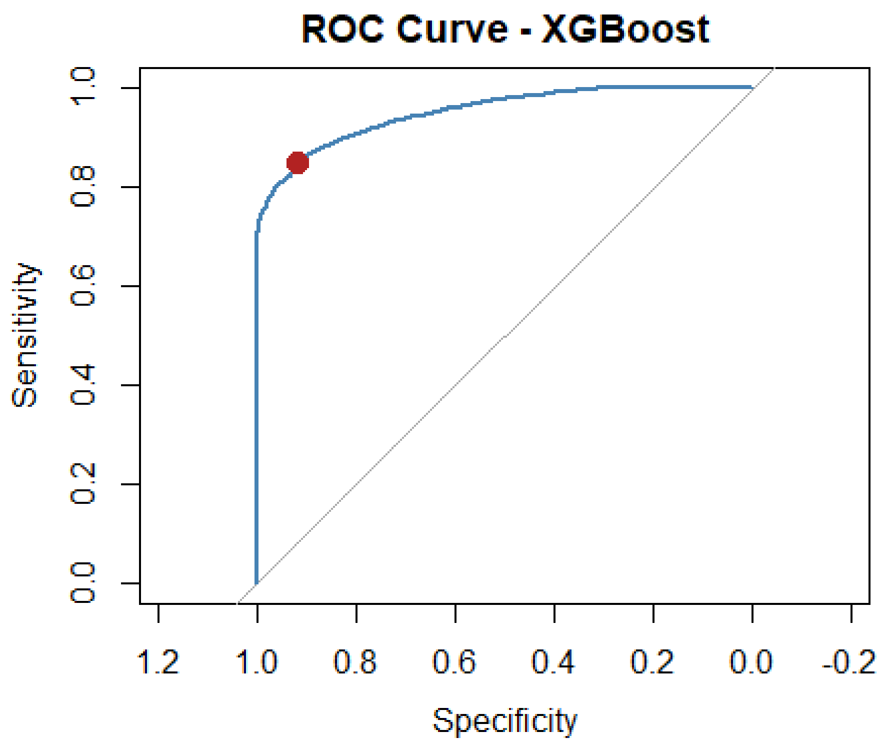
```r
plot(roc_obj, col = "steelblue", main = "ROC Curve - XGBoost")
```




```r
importance_matrix <- xgb.importance(model = xgb_model)
xgb.plot.importance(importance_matrix, top_n = 15, main = "Top 15 Features")
```



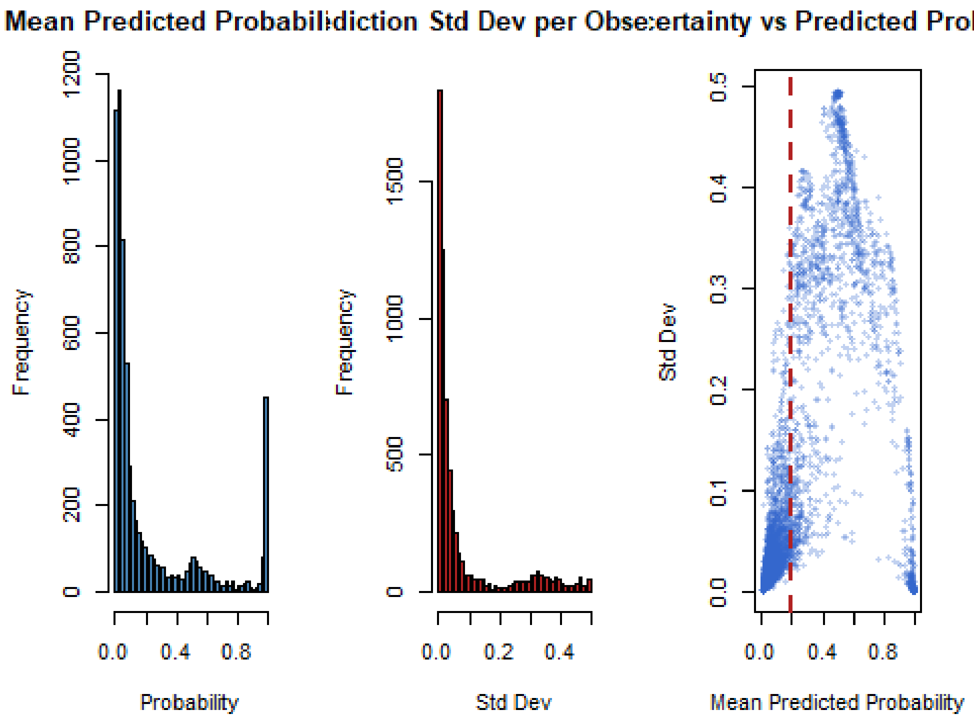

```r
roc_obj   <- roc(test_y, pred_prob)

## Setting levels: control = 0, case = 1

## Setting direction: controls < cases

youden    <- coords(roc_obj, "best", best.method = "youden")
print(youden)

##   threshold specificity sensitivity
## 1  0.192448   0.9176355   0.8460472

plot(roc_obj, col = "steelblue", main = "ROC Curve - XGBoost")
points(youden$specificity, youden$sensitivity, col = "firebrick", pch = 19,
cex = 1.5)
```




```
pred_class_optimal <- ifelse(pred_prob > 0.192, 1, 0)
confusionMatrix(factor(pred_class_optimal), factor(test_y), positive = "1")

## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##          0 4653  222
##          1  422 1220
##
##                Accuracy : 0.9012
##                  95% CI : (0.8937, 0.9083)
##     No Information Rate : 0.7787
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.7268
##
##  Mcnemar's Test P-Value : 4.445e-15
##
##             Sensitivity : 0.8460
##             Specificity : 0.9168
##          Pos Pred Value : 0.7430
##          Neg Pred Value : 0.9545
##              Prevalence : 0.2213
##          Detection Rate : 0.1872
##    Detection Prevalence : 0.2520
##       Balanced Accuracy : 0.8814
```

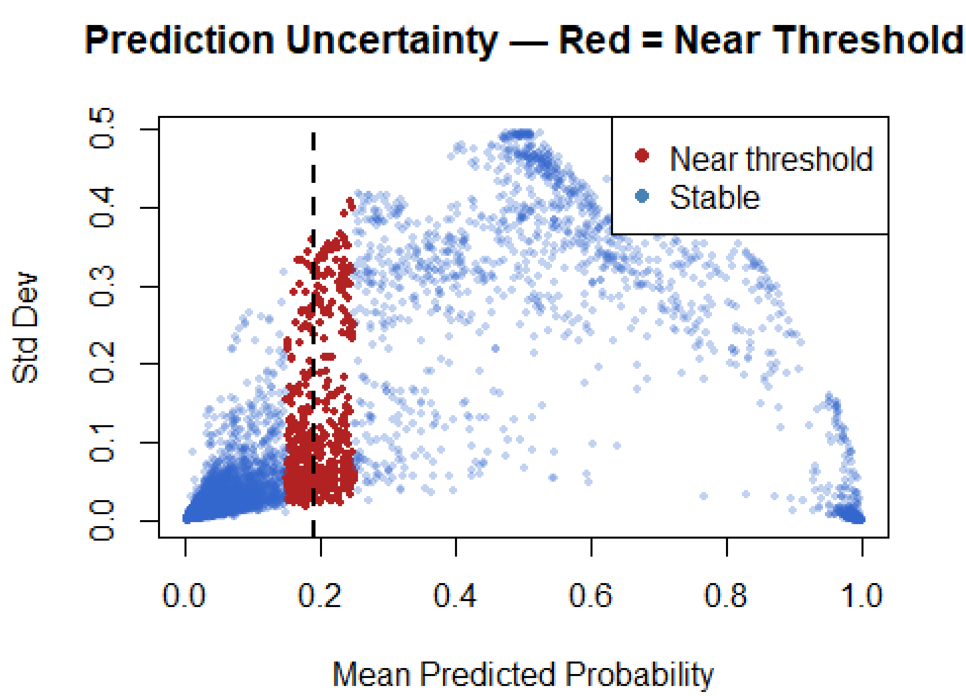
```
##
##        'Positive' Class : 1
##

set.seed(123)
n_simulations <- 1000
n_test        <- nrow(test_x)
mc_preds      <- matrix(NA, nrow = n_test, ncol = n_simulations)

for (i in 1:n_simulations) {

  noise          <- matrix(rnorm(n_test * ncol(test_x), mean = 0, sd = 0.01),
                           nrow = n_test, ncol = ncol(test_x))
  test_x_perturb <- test_x + noise

  d_perturb       <- xgb.DMatrix(data = test_x_perturb)
  mc_preds[, i]   <- predict(xgb_model, d_perturb)
}

mc_mean <- apply(mc_preds, 1, mean)
mc_sd   <- apply(mc_preds, 1, sd)
mc_cv   <- mc_sd / mc_mean

cat("Average prediction std dev:", round(mean(mc_sd), 4), "\n")

## Average prediction std dev: 0.082

cat("Max prediction std dev:", round(max(mc_sd), 4), "\n")

## Max prediction std dev: 0.4952

cat("Average coefficient of variation:", round(mean(mc_cv), 4), "\n")

## Average coefficient of variation: 0.5442

par(mfrow = c(1, 3))

hist(mc_mean, breaks = 50, col = "steelblue",
     main = "Mean Predicted Probabilities",
     xlab = "Probability")

hist(mc_sd, breaks = 50, col = "firebrick",
     main = "Prediction Std Dev per Observation",
     xlab = "Std Dev")

plot(mc_mean, mc_sd, col = rgb(0.2, 0.4, 0.8, 0.3),
     pch = 16, cex = 0.6,
     main = "Uncertainty vs Predicted Probability",
     xlab = "Mean Predicted Probability",
```

```r
     ylab = "Std Dev")
abline(v = 0.192, col = "firebrick", lty = 2, lwd = 2)
```




```r
par(mfrow = c(1, 1))

uncertainty_df <- data.frame(
  index     = 1:n_test,
  mean      = mc_mean,
  sd        = mc_sd,
  cv        = mc_cv,
  actual    = test_y
)

high_uncertainty <- uncertainty_df[mc_sd > quantile(mc_sd, 0.90), ]
cat("High uncertainty observations:", nrow(high_uncertainty), "\n")

## High uncertainty observations: 652

cat("Default rate in high uncertainty group:",
    round(mean(high_uncertainty$actual), 3), "\n")

## Default rate in high uncertainty group: 0.495

cat("Default rate in full test set:",
    round(mean(test_y), 3), "\n")

## Default rate in full test set: 0.221
```

```r
plot(mc_mean, mc_sd,
     col = ifelse(mc_mean > 0.15 & mc_mean < 0.25, "firebrick",
                  rgb(0.2, 0.4, 0.8, 0.3)),
     pch  = 16, cex = 0.6,
     main = "Prediction Uncertainty — Red = Near Threshold",
     xlab = "Mean Predicted Probability",
     ylab = "Std Dev")
abline(v = 0.192, col = "black", lty = 2, lwd = 2)
legend("topright", legend = c("Near threshold", "Stable"),
       col = c("firebrick", "steelblue"), pch = 16)
```




```r
zone <- sum(mc_mean > 0.15 & mc_mean < 0.25)
cat("Observations in threshold uncertainty zone:", zone, "\n")

## Observations in threshold uncertainty zone: 499

cat("As percentage of test set:", round(zone / n_test * 100, 2), "%\n")

## As percentage of test set: 7.66 %

zone_idx <- which(mc_mean > 0.15 & mc_mean < 0.25)
zone_df  <- data.frame(
  mean       = mc_mean[zone_idx],
  sd         = mc_sd[zone_idx],
  actual     = test_y[zone_idx]
)
```

```r
cat("Default rate in zone:", round(mean(zone_df$actual), 3), "\n")

## Default rate in zone: 0.17

cat("Avg std dev in zone:", round(mean(zone_df$sd), 3), "\n")

## Avg std dev in zone: 0.123

cat("Zone defaulters missed at 0.192:",
    sum(zone_df$actual == 1 & zone_df$mean < 0.192), "\n")

## Zone defaulters missed at 0.192: 36

cat("Zone good borrowers rejected at 0.192:",
    sum(zone_df$actual == 0 & zone_df$mean >= 0.192), "\n")

## Zone good borrowers rejected at 0.192: 189

decision <- ifelse(mc_mean < 0.15, "Approve",
            ifelse(mc_mean > 0.25, "Reject",
                   "Manual Review"))
table(decision)

## decision
##       Approve Manual Review        Reject
##          4353           499          1665

flag <- ifelse(mc_sd > quantile(mc_sd, 0.90), "Flag", "Auto")
table(flag)

## flag
## Auto Flag
## 5865  652

cv_result <- xgb.cv(
  params                = params,
  data                  = dtrain,
  nfold                 = 5,
  nrounds               = 500,
  early_stopping_rounds = 30,
  verbose               = 1,
  stratified            = TRUE
)

## Multiple eval metrics are present. Will use test_auc for early stopping.
## Will train until test_auc hasn't improved in 30 rounds.
##
## [1]  train-auc:0.895256±0.000682 test-auc:0.890416±0.003292
## [2]  train-auc:0.904274±0.003479 test-auc:0.901433±0.003509
## [3]  train-auc:0.913274±0.000576 test-auc:0.909987±0.002187
## [4]  train-auc:0.916667±0.000746 test-auc:0.913399±0.002131
## [5]  train-auc:0.917683±0.001725 test-auc:0.913765±0.002490
```

```
## [6]  train-auc:0.918244±0.001271 test-auc:0.913947±0.002421
## [7]  train-auc:0.919270±0.001461 test-auc:0.914924±0.002148
## [8]  train-auc:0.920438±0.001428 test-auc:0.915920±0.001753
## [9]  train-auc:0.921066±0.001306 test-auc:0.916535±0.002132
## [10] train-auc:0.921899±0.000648 test-auc:0.917129±0.002385
## [11] train-auc:0.922491±0.000540 test-auc:0.917243±0.002354
## [12] train-auc:0.923629±0.000463 test-auc:0.917745±0.001910
## [13] train-auc:0.924433±0.000514 test-auc:0.918332±0.001988
## [14] train-auc:0.925180±0.000556 test-auc:0.919200±0.002060
## [15] train-auc:0.925935±0.000528 test-auc:0.920001±0.002089
## [16] train-auc:0.926066±0.000633 test-auc:0.919962±0.002000
## [17] train-auc:0.926791±0.000686 test-auc:0.920602±0.001997
## [18] train-auc:0.926595±0.000652 test-auc:0.920412±0.001854
## [19] train-auc:0.926917±0.000630 test-auc:0.920807±0.001677
## [20] train-auc:0.928127±0.000699 test-auc:0.921913±0.001457
## [21] train-auc:0.928954±0.000787 test-auc:0.922778±0.001372
## [22] train-auc:0.929816±0.001113 test-auc:0.923655±0.001057
## [23] train-auc:0.930586±0.001196 test-auc:0.924257±0.000875
## [24] train-auc:0.931358±0.001183 test-auc:0.924978±0.001169
## [25] train-auc:0.932029±0.001089 test-auc:0.925323±0.001316
## [26] train-auc:0.932802±0.001152 test-auc:0.925861±0.001258
## [27] train-auc:0.933467±0.001126 test-auc:0.926378±0.000998
## [28] train-auc:0.934013±0.001092 test-auc:0.926973±0.000848
## [29] train-auc:0.934496±0.001113 test-auc:0.927308±0.000882
## [30] train-auc:0.935308±0.000961 test-auc:0.927628±0.000887
## [31] train-auc:0.935960±0.000975 test-auc:0.928067±0.000632
## [32] train-auc:0.936360±0.000865 test-auc:0.928295±0.000656
## [33] train-auc:0.936880±0.000750 test-auc:0.928558±0.000752
## [34] train-auc:0.937124±0.000676 test-auc:0.928834±0.000759
## [35] train-auc:0.937364±0.000567 test-auc:0.928992±0.000745
## [36] train-auc:0.937687±0.000605 test-auc:0.929348±0.000735
## [37] train-auc:0.938177±0.001067 test-auc:0.929720±0.001075
## [38] train-auc:0.938674±0.000988 test-auc:0.930042±0.001057
## [39] train-auc:0.939073±0.000792 test-auc:0.930290±0.001140
## [40] train-auc:0.939621±0.001027 test-auc:0.930841±0.000843
## [41] train-auc:0.939992±0.001066 test-auc:0.930997±0.000753
## [42] train-auc:0.940517±0.000999 test-auc:0.931453±0.000730
## [43] train-auc:0.941649±0.000832 test-auc:0.932329±0.000741
## [44] train-auc:0.942536±0.000790 test-auc:0.932982±0.000810
## [45] train-auc:0.943202±0.000548 test-auc:0.933459±0.000906
## [46] train-auc:0.943579±0.000603 test-auc:0.933718±0.000869
## [47] train-auc:0.943973±0.000500 test-auc:0.933847±0.000890
## [48] train-auc:0.944232±0.000510 test-auc:0.934007±0.000788
## [49] train-auc:0.944897±0.000803 test-auc:0.934520±0.000252
## [50] train-auc:0.945437±0.000825 test-auc:0.934910±0.000475
## [51] train-auc:0.945765±0.000907 test-auc:0.935138±0.000533
## [52] train-auc:0.946367±0.001186 test-auc:0.935744±0.001106
## [53] train-auc:0.946915±0.001126 test-auc:0.936108±0.000789
## [54] train-auc:0.947650±0.000966 test-auc:0.936771±0.000804
## [55] train-auc:0.948042±0.000900 test-auc:0.937105±0.000950
```

```
## [56] train-auc:0.948356±0.000880 test-auc:0.937348±0.000808
## [57] train-auc:0.948732±0.000872 test-auc:0.937378±0.000866
## [58] train-auc:0.949039±0.000798 test-auc:0.937561±0.000825
## [59] train-auc:0.949435±0.000706 test-auc:0.937664±0.000820
## [60] train-auc:0.949983±0.000750 test-auc:0.938052±0.001056
## [61] train-auc:0.950308±0.000846 test-auc:0.938213±0.001224
## [62] train-auc:0.950654±0.000751 test-auc:0.938503±0.001326
## [63] train-auc:0.951172±0.000758 test-auc:0.938942±0.001167
## [64] train-auc:0.951737±0.000881 test-auc:0.939203±0.001095
## [65] train-auc:0.952122±0.001011 test-auc:0.939421±0.001180
## [66] train-auc:0.952600±0.000797 test-auc:0.939726±0.001244
## [67] train-auc:0.952977±0.000914 test-auc:0.939939±0.001393
## [68] train-auc:0.953470±0.000748 test-auc:0.940159±0.001486
## [69] train-auc:0.953882±0.000756 test-auc:0.940155±0.001496
## [70] train-auc:0.954328±0.000833 test-auc:0.940314±0.001389
## [71] train-auc:0.954735±0.000719 test-auc:0.940475±0.001292
## [72] train-auc:0.955040±0.000496 test-auc:0.940625±0.001287
## [73] train-auc:0.955365±0.000442 test-auc:0.940648±0.001357
## [74] train-auc:0.955687±0.000476 test-auc:0.940801±0.001376
## [75] train-auc:0.955962±0.000503 test-auc:0.940792±0.001388
## [76] train-auc:0.956394±0.000650 test-auc:0.940954±0.001192
## [77] train-auc:0.956787±0.000739 test-auc:0.940943±0.001133
## [78] train-auc:0.957114±0.000832 test-auc:0.941070±0.001146
## [79] train-auc:0.957465±0.000924 test-auc:0.941358±0.001073
## [80] train-auc:0.957956±0.000886 test-auc:0.941772±0.000926
## [81] train-auc:0.958260±0.000911 test-auc:0.941909±0.000945
## [82] train-auc:0.958540±0.000868 test-auc:0.941981±0.001027
## [83] train-auc:0.958809±0.000889 test-auc:0.941996±0.001077
## [84] train-auc:0.959158±0.000999 test-auc:0.942202±0.001110
## [85] train-auc:0.959420±0.001042 test-auc:0.942266±0.001079
## [86] train-auc:0.959686±0.001024 test-auc:0.942351±0.001073
## [87] train-auc:0.960105±0.001040 test-auc:0.942384±0.001161
## [88] train-auc:0.960317±0.001064 test-auc:0.942422±0.001186
## [89] train-auc:0.960675±0.001118 test-auc:0.942639±0.001064
## [90] train-auc:0.960964±0.001123 test-auc:0.942740±0.001067
## [91] train-auc:0.961360±0.001153 test-auc:0.942954±0.001004
## [92] train-auc:0.961661±0.001134 test-auc:0.942931±0.001110
## [93] train-auc:0.961888±0.001135 test-auc:0.942995±0.001052
## [94] train-auc:0.962157±0.001085 test-auc:0.943056±0.001075
## [95] train-auc:0.962528±0.001089 test-auc:0.943072±0.001084
## [96] train-auc:0.962888±0.001143 test-auc:0.943283±0.001109
## [97] train-auc:0.963081±0.001064 test-auc:0.943305±0.001226
## [98] train-auc:0.963242±0.001006 test-auc:0.943408±0.001203
## [99] train-auc:0.963556±0.000980 test-auc:0.943555±0.001260
## [100]    train-auc:0.963797±0.001028 test-auc:0.943653±0.001249
## [101]    train-auc:0.964001±0.001055 test-auc:0.943689±0.001190
## [102]    train-auc:0.964169±0.001101 test-auc:0.943766±0.001142
## [103]    train-auc:0.964461±0.001098 test-auc:0.943914±0.001236
## [104]    train-auc:0.964667±0.001129 test-auc:0.944077±0.001334
## [105]    train-auc:0.965009±0.001112 test-auc:0.944210±0.001408
```

```
## [106]	train-auc:0.965371±0.001195 test-auc:0.944192±0.001478
## [107]	train-auc:0.965640±0.001278 test-auc:0.944305±0.001575
## [108]	train-auc:0.965764±0.001200 test-auc:0.944356±0.001584
## [109]	train-auc:0.966033±0.001169 test-auc:0.944428±0.001474
## [110]	train-auc:0.966333±0.001116 test-auc:0.944672±0.001575
## [111]	train-auc:0.966526±0.001031 test-auc:0.944728±0.001730
## [112]	train-auc:0.966687±0.001061 test-auc:0.944848±0.001692
## [113]	train-auc:0.966957±0.001023 test-auc:0.944922±0.001701
## [114]	train-auc:0.967072±0.001054 test-auc:0.944962±0.001715
## [115]	train-auc:0.967215±0.001047 test-auc:0.944999±0.001645
## [116]	train-auc:0.967391±0.001003 test-auc:0.945000±0.001666
## [117]	train-auc:0.967639±0.000904 test-auc:0.945120±0.001789
## [118]	train-auc:0.967866±0.000967 test-auc:0.945147±0.001920
## [119]	train-auc:0.968097±0.001026 test-auc:0.945233±0.001938
## [120]	train-auc:0.968305±0.000983 test-auc:0.945236±0.001971
## [121]	train-auc:0.968581±0.000887 test-auc:0.945376±0.001925
## [122]	train-auc:0.968724±0.000835 test-auc:0.945484±0.001952
## [123]	train-auc:0.968943±0.000792 test-auc:0.945590±0.001974
## [124]	train-auc:0.969082±0.000842 test-auc:0.945663±0.001998
## [125]	train-auc:0.969287±0.000785 test-auc:0.945663±0.002038
## [126]	train-auc:0.969599±0.000735 test-auc:0.945695±0.002039
## [127]	train-auc:0.969748±0.000805 test-auc:0.945627±0.002111
## [128]	train-auc:0.969912±0.000749 test-auc:0.945668±0.002143
## [129]	train-auc:0.970090±0.000790 test-auc:0.945692±0.002062
## [130]	train-auc:0.970296±0.000800 test-auc:0.945732±0.002106
## [131]	train-auc:0.970492±0.000772 test-auc:0.945825±0.002060
## [132]	train-auc:0.970626±0.000848 test-auc:0.945785±0.002032
## [133]	train-auc:0.970743±0.000760 test-auc:0.945797±0.002063
## [134]	train-auc:0.970926±0.000804 test-auc:0.945773±0.002095
## [135]	train-auc:0.971124±0.000816 test-auc:0.945814±0.002043
## [136]	train-auc:0.971245±0.000864 test-auc:0.945781±0.002069
## [137]	train-auc:0.971467±0.000902 test-auc:0.945747±0.002057
## [138]	train-auc:0.971624±0.000892 test-auc:0.945815±0.002111
## [139]	train-auc:0.971795±0.000886 test-auc:0.945764±0.002072
## [140]	train-auc:0.971899±0.000854 test-auc:0.945763±0.002115
## [141]	train-auc:0.972067±0.000941 test-auc:0.945789±0.002136
## [142]	train-auc:0.972205±0.000939 test-auc:0.945869±0.002098
## [143]	train-auc:0.972309±0.000977 test-auc:0.945884±0.002141
## [144]	train-auc:0.972511±0.000995 test-auc:0.945895±0.002137
## [145]	train-auc:0.972684±0.001015 test-auc:0.945879±0.002187
## [146]	train-auc:0.972832±0.001091 test-auc:0.945894±0.002136
## [147]	train-auc:0.972970±0.001096 test-auc:0.945909±0.002077
## [148]	train-auc:0.973152±0.001103 test-auc:0.945941±0.002163
## [149]	train-auc:0.973235±0.001166 test-auc:0.946012±0.002176
## [150]	train-auc:0.973510±0.001154 test-auc:0.946040±0.002139
## [151]	train-auc:0.973565±0.001141 test-auc:0.946027±0.002203
## [152]	train-auc:0.973704±0.001104 test-auc:0.946067±0.002251
## [153]	train-auc:0.973844±0.001150 test-auc:0.946147±0.002279
## [154]	train-auc:0.974041±0.001138 test-auc:0.946159±0.002365
## [155]	train-auc:0.974207±0.001086 test-auc:0.946094±0.002330
```

```
## [156]	train-auc:0.974319±0.001051 test-auc:0.946154±0.002311
## [157]	train-auc:0.974449±0.001084 test-auc:0.946175±0.002316
## [158]	train-auc:0.974561±0.001067 test-auc:0.946254±0.002286
## [159]	train-auc:0.974647±0.001104 test-auc:0.946289±0.002348
## [160]	train-auc:0.974705±0.001146 test-auc:0.946310±0.002349
## [161]	train-auc:0.974832±0.001189 test-auc:0.946367±0.002396
## [162]	train-auc:0.975000±0.001277 test-auc:0.946330±0.002384
## [163]	train-auc:0.975125±0.001248 test-auc:0.946395±0.002398
## [164]	train-auc:0.975270±0.001253 test-auc:0.946381±0.002405
## [165]	train-auc:0.975455±0.001224 test-auc:0.946391±0.002359
## [166]	train-auc:0.975578±0.001219 test-auc:0.946375±0.002370
## [167]	train-auc:0.975685±0.001198 test-auc:0.946422±0.002497
## [168]	train-auc:0.975852±0.001217 test-auc:0.946481±0.002494
## [169]	train-auc:0.975918±0.001266 test-auc:0.946531±0.002530
## [170]	train-auc:0.976085±0.001249 test-auc:0.946519±0.002533
## [171]	train-auc:0.976227±0.001330 test-auc:0.946513±0.002551
## [172]	train-auc:0.976300±0.001307 test-auc:0.946518±0.002559
## [173]	train-auc:0.976452±0.001334 test-auc:0.946443±0.002567
## [174]	train-auc:0.976512±0.001366 test-auc:0.946460±0.002571
## [175]	train-auc:0.976623±0.001319 test-auc:0.946443±0.002611
## [176]	train-auc:0.976723±0.001264 test-auc:0.946470±0.002597
## [177]	train-auc:0.976874±0.001317 test-auc:0.946466±0.002544
## [178]	train-auc:0.976957±0.001308 test-auc:0.946531±0.002652
## [179]	train-auc:0.977030±0.001333 test-auc:0.946585±0.002621
## [180]	train-auc:0.977105±0.001277 test-auc:0.946626±0.002660
## [181]	train-auc:0.977169±0.001268 test-auc:0.946649±0.002658
## [182]	train-auc:0.977229±0.001309 test-auc:0.946656±0.002619
## [183]	train-auc:0.977308±0.001240 test-auc:0.946655±0.002582
## [184]	train-auc:0.977412±0.001248 test-auc:0.946689±0.002555
## [185]	train-auc:0.977489±0.001279 test-auc:0.946651±0.002603
## [186]	train-auc:0.977573±0.001243 test-auc:0.946662±0.002632
## [187]	train-auc:0.977648±0.001265 test-auc:0.946698±0.002646
## [188]	train-auc:0.977729±0.001245 test-auc:0.946676±0.002673
## [189]	train-auc:0.977872±0.001221 test-auc:0.946800±0.002656
## [190]	train-auc:0.977973±0.001120 test-auc:0.946794±0.002651
## [191]	train-auc:0.978097±0.001118 test-auc:0.946759±0.002695
## [192]	train-auc:0.978169±0.001123 test-auc:0.946796±0.002700
## [193]	train-auc:0.978249±0.001067 test-auc:0.946773±0.002756
## [194]	train-auc:0.978433±0.001042 test-auc:0.946887±0.002738
## [195]	train-auc:0.978528±0.001063 test-auc:0.946889±0.002763
## [196]	train-auc:0.978649±0.001073 test-auc:0.946988±0.002778
## [197]	train-auc:0.978731±0.001089 test-auc:0.947010±0.002708
## [198]	train-auc:0.978848±0.001074 test-auc:0.947004±0.002686
## [199]	train-auc:0.978929±0.001078 test-auc:0.947030±0.002718
## [200]	train-auc:0.978971±0.001041 test-auc:0.947044±0.002751
## [201]	train-auc:0.979097±0.001025 test-auc:0.947138±0.002741
## [202]	train-auc:0.979139±0.001013 test-auc:0.947121±0.002730
## [203]	train-auc:0.979194±0.001038 test-auc:0.947135±0.002728
## [204]	train-auc:0.979276±0.001020 test-auc:0.947098±0.002686
## [205]	train-auc:0.979362±0.001039 test-auc:0.947099±0.002771
```

```
## [206]	train-auc:0.979411±0.001058 test-auc:0.947127±0.002803
## [207]	train-auc:0.979481±0.001060 test-auc:0.947117±0.002817
## [208]	train-auc:0.979586±0.001090 test-auc:0.947168±0.002822
## [209]	train-auc:0.979632±0.001066 test-auc:0.947205±0.002897
## [210]	train-auc:0.979689±0.001090 test-auc:0.947178±0.002902
## [211]	train-auc:0.979759±0.001072 test-auc:0.947158±0.002905
## [212]	train-auc:0.979810±0.001077 test-auc:0.947198±0.002868
## [213]	train-auc:0.979941±0.001092 test-auc:0.947232±0.002790
## [214]	train-auc:0.980026±0.001088 test-auc:0.947222±0.002834
## [215]	train-auc:0.980107±0.001071 test-auc:0.947169±0.002808
## [216]	train-auc:0.980162±0.001085 test-auc:0.947221±0.002820
## [217]	train-auc:0.980202±0.001064 test-auc:0.947188±0.002774
## [218]	train-auc:0.980258±0.001027 test-auc:0.947193±0.002786
## [219]	train-auc:0.980309±0.000992 test-auc:0.947201±0.002750
## [220]	train-auc:0.980383±0.001004 test-auc:0.947200±0.002735
## [221]	train-auc:0.980441±0.000981 test-auc:0.947200±0.002830
## [222]	train-auc:0.980524±0.000963 test-auc:0.947155±0.002812
## [223]	train-auc:0.980546±0.000933 test-auc:0.947180±0.002830
## [224]	train-auc:0.980687±0.000976 test-auc:0.947215±0.002737
## [225]	train-auc:0.980792±0.000991 test-auc:0.947240±0.002773
## [226]	train-auc:0.980825±0.000990 test-auc:0.947283±0.002830
## [227]	train-auc:0.980879±0.001005 test-auc:0.947218±0.002905
## [228]	train-auc:0.980927±0.001001 test-auc:0.947220±0.002926
## [229]	train-auc:0.981011±0.001070 test-auc:0.947224±0.002932
## [230]	train-auc:0.981080±0.001068 test-auc:0.947241±0.002952
## [231]	train-auc:0.981186±0.001068 test-auc:0.947242±0.002964
## [232]	train-auc:0.981244±0.001086 test-auc:0.947270±0.002988
## [233]	train-auc:0.981278±0.001107 test-auc:0.947263±0.003024
## [234]	train-auc:0.981353±0.001120 test-auc:0.947222±0.002998
## [235]	train-auc:0.981422±0.001153 test-auc:0.947230±0.002974
## [236]	train-auc:0.981482±0.001132 test-auc:0.947212±0.003037
## [237]	train-auc:0.981566±0.001152 test-auc:0.947253±0.002974
## [238]	train-auc:0.981620±0.001125 test-auc:0.947301±0.002916
## [239]	train-auc:0.981700±0.001098 test-auc:0.947253±0.002898
## [240]	train-auc:0.981761±0.001108 test-auc:0.947280±0.002914
## [241]	train-auc:0.981846±0.001042 test-auc:0.947325±0.002855
## [242]	train-auc:0.981905±0.001074 test-auc:0.947350±0.002875
## [243]	train-auc:0.981976±0.001044 test-auc:0.947381±0.002858
## [244]	train-auc:0.982047±0.001034 test-auc:0.947419±0.002914
## [245]	train-auc:0.982108±0.001001 test-auc:0.947390±0.002878
## [246]	train-auc:0.982158±0.001043 test-auc:0.947381±0.002884
## [247]	train-auc:0.982188±0.001066 test-auc:0.947386±0.002885
## [248]	train-auc:0.982210±0.001088 test-auc:0.947378±0.002898
## [249]	train-auc:0.982307±0.001065 test-auc:0.947390±0.002906
## [250]	train-auc:0.982379±0.001059 test-auc:0.947296±0.002935
## [251]	train-auc:0.982437±0.001031 test-auc:0.947292±0.002977
## [252]	train-auc:0.982487±0.001027 test-auc:0.947303±0.002959
## [253]	train-auc:0.982523±0.001006 test-auc:0.947297±0.002965
## [254]	train-auc:0.982587±0.001021 test-auc:0.947303±0.002988
## [255]	train-auc:0.982651±0.001065 test-auc:0.947290±0.002988
```

```
## [256]	train-auc:0.982715±0.001056 test-auc:0.947279±0.003033
## [257]	train-auc:0.982760±0.001040 test-auc:0.947252±0.003057
## [258]	train-auc:0.982841±0.001007 test-auc:0.947261±0.003054
## [259]	train-auc:0.982876±0.001030 test-auc:0.947309±0.003004
## [260]	train-auc:0.982943±0.000992 test-auc:0.947274±0.003047
## [261]	train-auc:0.983023±0.000997 test-auc:0.947291±0.003009
## [262]	train-auc:0.983110±0.001018 test-auc:0.947336±0.002986
## [263]	train-auc:0.983124±0.001024 test-auc:0.947360±0.002985
## [264]	train-auc:0.983145±0.001036 test-auc:0.947361±0.002964
## [265]	train-auc:0.983212±0.001097 test-auc:0.947395±0.002989
## [266]	train-auc:0.983259±0.001085 test-auc:0.947364±0.003008
## [267]	train-auc:0.983330±0.001036 test-auc:0.947357±0.003118
## [268]	train-auc:0.983386±0.001071 test-auc:0.947309±0.003188
## [269]	train-auc:0.983430±0.001066 test-auc:0.947316±0.003186
## [270]	train-auc:0.983458±0.001086 test-auc:0.947336±0.003171
## [271]	train-auc:0.983502±0.001062 test-auc:0.947346±0.003138
## [272]	train-auc:0.983570±0.001037 test-auc:0.947357±0.003132
## [273]	train-auc:0.983622±0.001065 test-auc:0.947348±0.003128
## Stopping. Best iteration:
## [274]	train-auc:0.983717±0.001020 test-auc:0.947305±0.003190
##
## [274]	train-auc:0.983717±0.001020 test-auc:0.947305±0.003190
```

```r
# results
best_iter <- cv_result$best_iteration
cat("Best iteration:", best_iter, "\n")
```

```
## Best iteration:
```

```r
cat("CV AUC mean:", round(cv_result$evaluation_log$test_auc_mean[best_iter],
4), "\n")
```

```
## CV AUC mean:
```

```r
cat("CV AUC std dev:",
round(cv_result$evaluation_log$test_auc_std[best_iter], 4), "\n")
```

```
## CV AUC std dev:
```

```r
cat("95% CI: [",
    round(cv_result$evaluation_log$test_auc_mean[best_iter] -
          1.96 * cv_result$evaluation_log$test_auc_std[best_iter], 4), ",",
    round(cv_result$evaluation_log$test_auc_mean[best_iter] +
          1.96 * cv_result$evaluation_log$test_auc_std[best_iter], 4), "]\n")
```

```
## 95% CI: [  ,  ]
```

```r
# best iteration from CV
best_iter <- which.max(cv_result$evaluation_log$test_auc_mean)
cat("Best iteration:", best_iter, "\n")
```

```
## Best iteration: 244
```

```r
cat("CV AUC mean:", round(cv_result$evaluation_log$test_auc_mean[best_iter],
4), "\n")

## CV AUC mean: 0.9474

cat("CV AUC std dev:",
round(cv_result$evaluation_log$test_auc_std[best_iter], 4), "\n")

## CV AUC std dev: 0.0029

cat("95% CI: [",
    round(cv_result$evaluation_log$test_auc_mean[best_iter] -
          1.96 * cv_result$evaluation_log$test_auc_std[best_iter], 4), ",",
    round(cv_result$evaluation_log$test_auc_mean[best_iter] +
          1.96 * cv_result$evaluation_log$test_auc_std[best_iter], 4), "]\n")

## 95% CI: [ 0.9417 , 0.9531 ]

# train final model using best_iter from CV instead of early stopping
xgb_model_final <- xgb.train(
  params  = params,
  data    = dtrain,
  nrounds = best_iter
)

# evaluate on test set
pred_prob_final  <- predict(xgb_model_final, dtest)
pred_class_final <- ifelse(pred_prob_final > 0.192, 1, 0)

confusionMatrix(factor(pred_class_final), factor(test_y), positive = "1")

## Confusion Matrix and Statistics
##
##             Reference
## Prediction    0    1
##          0 4660  230
##          1  415 1212
##
##                Accuracy : 0.901
##                  95% CI : (0.8935, 0.9082)
##     No Information Rate : 0.7787
##     P-Value [Acc > NIR] : < 2.2e-16
##
##                   Kappa : 0.7254
##
##  Mcnemar's Test P-Value : 4.325e-13
##
##             Sensitivity : 0.8405
##             Specificity : 0.9182
##          Pos Pred Value : 0.7449
##          Neg Pred Value : 0.9530
```

```
##               Prevalence : 0.2213
##           Detection Rate : 0.1860
##     Detection Prevalence : 0.2497
##        Balanced Accuracy : 0.8794
##
##         'Positive' Class : 1
##
```

```r
roc_obj_final <- roc(test_y, pred_prob_final)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```r
cat("Final AUC:", auc(roc_obj_final), "\n")
```

```
## Final AUC: 0.9494372
```

```r
cat("CV AUC mean:", round(cv_result$evaluation_log$test_auc_mean[best_iter],
4), "\n")
```

```
## CV AUC mean: 0.9474
```

```r
cat("CV AUC std dev:",
round(cv_result$evaluation_log$test_auc_std[best_iter], 4), "\n")
```

```
## CV AUC std dev: 0.0029
```

```r
cat("95% CI: [",
    round(cv_result$evaluation_log$test_auc_mean[best_iter] -
          1.96 * cv_result$evaluation_log$test_auc_std[best_iter], 4), ",",
    round(cv_result$evaluation_log$test_auc_mean[best_iter] +
          1.96 * cv_result$evaluation_log$test_auc_std[best_iter], 4), "]\n")
```

```
## 95% CI: [ 0.9417 , 0.9531 ]
```

```r
# create calibration data
cal_df <- data.frame(
  pred_prob = pred_prob_final,
  actual    = test_y
)

# bin predictions into deciles
cal_df$bin <- cut(cal_df$pred_prob,
                  breaks = seq(0, 1, by = 0.1),
                  include.lowest = TRUE)

# calculate mean predicted vs mean actual per bin
cal_summary <- aggregate(cbind(pred_prob, actual) ~ bin,
                         data = cal_df,
                         FUN = mean)
```

```r
# plot
plot(cal_summary$pred_prob, cal_summary$actual,
     xlim = c(0, 1), ylim = c(0, 1),
     xlab = "Mean Predicted Probability",
     ylab = "Mean Actual Default Rate",
     main = "Calibration Plot — XGBoost",
     pch  = 19, col = "steelblue", cex = 1.5)
abline(0, 1, col = "firebrick", lty = 2, lwd = 2)
legend("topleft", legend = c("Model", "Perfect Calibration"),
       col = c("steelblue", "firebrick"), pch = c(19, NA), lty = c(NA, 2))
```